

MATH310 Probability & Statistics

Section L2

Week 2: Conditional Probability

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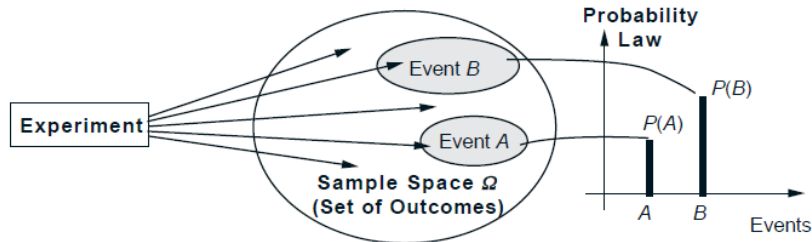
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Probabilistic Model

Elements of a Probabilistic Model

- The sample space Ω , which is the set of all possible outcomes of an experiment.
- The probability law, which assigns to a set A of possible outcomes (also called an event) a nonnegative number $P(A)$ (called the probability of A) that encodes our knowledge or belief about the collective “likelihood” of the elements of A . The probability law must satisfy certain properties to be introduced shortly.



Probability laws

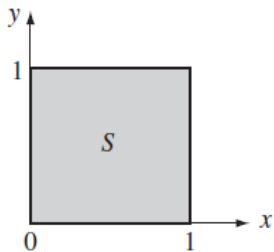
- ▶ The probability law assigns to an event E a non-negative number $P(E)$ which encodes our belief/knowledge about the “likelihood” of the event E .
- ▶ Axioms of probability:
 - ▶ **Nonnegativity:** $P(A) \geq 0$, for every event A .
 - ▶ **Additivity:** If A and B are two disjoint events, then the probability of their union satisfies $P(A \cup B) = P(A) + P(B)$.
This extends to the union of infinitely many disjoint events:

$$P(A_1 \cup A_2 \cup \dots) = P(A_1) + P(A_2) + \dots$$

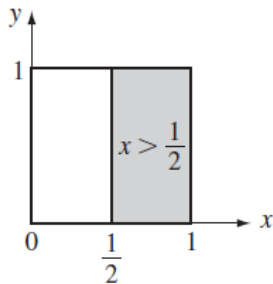
- ▶ **Normalization:** The probability of the entire sample space Ω is equal to 1, i.e. $P(\Omega) = 1$

Example

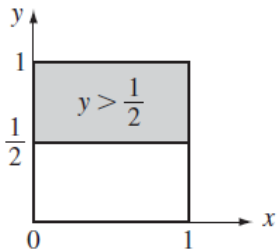
Consider an Experiment where we picked two numbers x and y at random between zero and one. The sample space is then the unit square.. If we suppose that all pairs of numbers in the unit square are equally likely to be selected, then it is reasonable to use a probability assignment in which the probability of any region R inside the unit square is equal to the area of R . Find the probability of the following events: $A = \{ x > 0.5 \}$ and $B = \{ y > 0.5 \}$ $C = \{ x > y \}$.



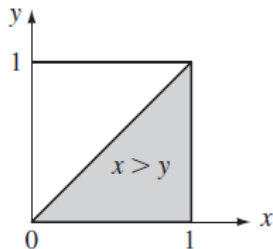
(a) Sample space



(b) Event $\{x > \frac{1}{2}\}$



(c) Event $\{y > \frac{1}{2}\}$



(d) Event $\{x > y\}$

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Alice and Bob toss two fair coins separately. Denote the event that Alice gets a H by E_A and the event that Bob gets a H by E_B . We know that $P(E_A) = P(E_B) = 1/2$.

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- ▶ May be what you are really thinking about is independent and not disjoint events.

Partial information

- ▶ So far, we have assumed we know nothing about the outcome of our experiment, except for the information encoded in the probability law.
- ▶ Sometimes, however, we have **partial information** that may affect the likelihood of a given event.
 - ▶ The experiment involves rolling a die. You are told that the number is odd.
 - ▶ The experiment involves the weather tomorrow. You know that the weather today is rainy.
 - ▶ The experiment involves the presence or absence of a disease. A blood test comes back positive.
- ▶ In each case, knowing about some event B (e.g. “it is raining today”) changes our beliefs about event A (“Will it rain tomorrow?”).
- ▶ We want to **update** our probability law to incorporate this new knowledge.

Conditional Probability

Original problem:

- ▶ What is the probability of some event A .
 - ▶ e.g. What is the probability that we roll a number less than 4?
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New problem:

- ▶ Assuming event B (equivalently given event B), what is the probability of event A ?
 - ▶ e.g. Given that the number rolled is an odd number, what is the probability that it is less than 4?
- ▶ We call this the **conditional distribution** of A given B .
- ▶ We write this as $P(A|B)$
- ▶ Read $|$ as “given” or “conditioned on the fact that”.
- ▶ Our conditional probability is still describing “the probability of something”, so we expect it to behave like a probability distribution.

Conditional Probability

- ▶ Consider rolling a fair 6-sided die (uniform, discrete probability distribution).
- ▶ Let A be the event “outcome is equal to 1’.
 - ▶ What is $P(A)$?
- ▶ Let’s now assume that the number rolled is an odd number.
 - ▶ What is the set, B , that we are conditioning on?
- ▶ What do you think $P(A|B)$ should be?

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- ▶ More generally, $P(A|B) = \frac{P(A \cap B)}{P(B)}$.
- ▶ A conditional probability is only defined if $P(B) > 0$.

Conditional Probability Axioms

- ▶ **Nonnegativity** – Check.
- ▶ **Normalization** –
- ▶ **Additivity** –

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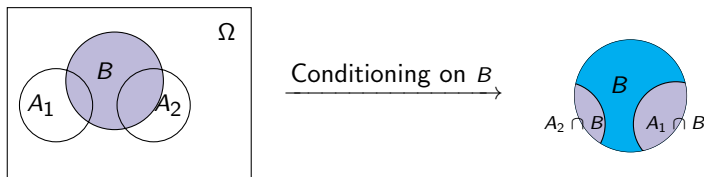
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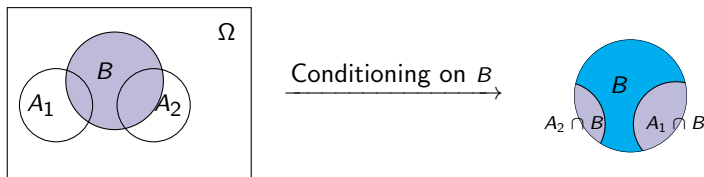
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Using additivity, $P((A_1 \cup A_2) \cap B) = P(A_1 \cap B) + P(A_2 \cap B)$, so

$$P(A_1 \cup A_2|B) = \frac{P(A_1 \cap B) + P(A_2 \cap B)}{P(B)} = P(A_1|B) + P(A_2|B)$$

Properties of conditional probability

If $P(B) > 0$,

- ▶ If A_i for $i \in \{1, \dots, n\}$ are all pairwise disjoint, then

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Example: Coin toss

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